

Thanh Phung Truong

Ph.D. Candidate, Computer Science and Engineering

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Research Profile

Ph.D. candidate in Computer Science and Engineering at Chung-Ang University, advised by Prof. Sungrae Cho. Research focuses on wireless communication, applied machine learning, and network architectures, with work spanning NOMA/RSMA, RIS/IRS-aided systems, semantic communications, quantized MIMO, UAV/HAP-enabled and maritime networks, digital twin edge networks, mobile edge computing, federated learning, and resource allocation for 5G/6G communication systems.

Education

Ph.D. in Computer Science and Engineering, Chung-Ang University, Seoul, South Korea Mar. 2023–Present

Advisor: Prof. Sungrae Cho

M.E. in Computer Science and Engineering, Chung-Ang University, Seoul, South Korea 2022

B.E. in Electronics and Telecommunications, Ho Chi Minh City University of Technology, Vietnam National University Ho Chi Minh City, Vietnam 2018

Research and Professional Experience

Research Assistant, Ultra-Intelligent Computing/Communication Laboratory, Chung-Ang University 2023–Present

Data Scientist, Viettel Networks, Viettel Group, Vietnam 2022–2023

FPGA Engineer, VinSmart Research and Manufacture Joint Stock Company, Vingroup, Vietnam 2019–2020

Research Interests

- Wireless communications and 5G/6G networks: NOMA, RSMA, semantic communications, quantized MIMO, RIS/IRS, energy efficiency, and resource allocation.
- AI and machine learning for networks: deep reinforcement learning, federated learning, semantic communication architectures, and intelligent resource management.
- Network architectures: UAV/HAP-assisted networks, maritime IoT, mobile edge computing, digital twin edge networks, and quantum communication.

Honors, Scholarships, and Awards

Scholarships

- **CAU Alumni Scholarship**, Chung-Ang University Graduate School Alumni Association, 2026.
- **Chung-Ang University Young Scientist Scholarship**, 2023.
- **Chung-Ang University Young Scientist Scholarship**, 2020.

Research Excellence

- **Excellence in Research Award 2024**, School of Computer Science and Engineering, Chung-Ang University; doctoral-course top distinction and paper-of-the-year recognition, awarded May 28, 2025.
- **Excellence in Research Award 2023**, School of Computer Science and Engineering, Chung-Ang University; doctoral-course top distinction and paper-of-the-year recognition, awarded June 24, 2024.

Paper Awards

- **Best Paper Award**, ICOIN 2026, “Multi-Task Semantic Communication System for DNA Transmission.”

- **Excellent Paper Award**, ICMIC 2025, “Efficient Offloading for Edge Computing in UAV-Assisted Maritime Networks.”
- **Excellent Paper Award**, ICAIIC 2025, “Super-Resolution Semantic Communication System for Satellite Image.”
- **Best Paper Award**, ICIT 2023, “Deep Reinforcement Learning-Based Sum-Rate Maximization for Uplink Multi-user SIMO-RSMA Systems.”
- **Outstanding Paper Award**, ICMIC 2023, “Multiple Access in HAP-Enabled Maritime IoT Networks: A Deep Reinforcement Learning Approach.”
- **Best Paper Award**, ICTC 2022, “Deep Reinforcement Learning-based Partial Task Offloading in High Altitude Platform-aided Vehicular Networks.”

Publications

Journal Articles

1. **T. P. Truong**, N.-P. Tran, J. Oh, D. Lee, V. D. Tuong, N.-N. Dao, and S. Cho, “Energy Efficiency in Quantized MIMO-NOMA Communication Systems,” *IEEE Transactions on Cognitive Communications and Networking*, vol. 12, pp. 6629–6642, March 2026, doi: 10.1109/TCCN.2024.3512847.
2. A.-T. Tran, **T. P. Truong**, D. Won, N.-N. Dao, and S. Cho, “Weighted Sum-Rate Maximization in Rate-Splitting MISO Downlink Systems,” *IEEE Transactions on Network Science and Engineering*, vol. 13, pp. 5522–5538, January 2026, doi: 10.1109/TNSE.2024.3498735.
3. T. S. Do, **T. P. Truong**, Q. T. Do, D. Won, W. Noh, and S. Cho, “BidDeepSC-1.58b: 1.58-bit Bidirectional Slimmable Semantic Communication System,” *IEEE Transactions on Vehicular Technology*, to appear, 2026, doi: 10.1109/TVT.2024.3511426.
4. T. H. Moges, **T. P. Truong**, D. S. Lakew, T. H. Huong, V. T. Hoang, N.-N. Dao, and S. Cho, “Age of Information-Aware Trajectory Optimization for Time-Sensitive UAV Systems in Uplink SCMA Networks,” *Computer Networks*, vol. 268, no. 111369, August 2025.
5. T. S. Do, **T. P. Truong**, Q. T. Do, and S. Cho, “TranGDeepSC: Leveraging ViT Knowledge in CNN-based Semantic Communication System,” *ICT Express*, vol. 11, no. 2, pp. 335–340, April 2025.
6. **T. P. Truong**, T. M. T. Nguyen, T. V. Nguyen, N.-N. Dao, and S. Cho, “Energy Efficiency in RSMA-Enhanced Active RIS-Aided Quantized Downlink Systems,” *IEEE Journal on Selected Areas in Communications*, vol. 43, no. 3, March 2025, doi: 10.1109/JSAC.2024.3521406.
7. **T. P. Truong**, H. V. Nguyen, N.-N. Dao, W. Noh, and S. Cho, “Orthogonalized RSMA based Flexible Multiple Access in Digital Twin Edge Networks,” *IEEE Transactions on Wireless Communications*, vol. 23, no. 12, December 2024, doi: 10.1109/TWC.2024.3468791.
8. T.-H. Nguyen, **T. P. Truong**, A.-T. Tran, N.-N. Dao, L. Park, and S. Cho, “Intelligent Heterogeneous Aerial Edge Computing for Advanced 5G Access,” *IEEE Transactions on Network Science and Engineering*, vol. 11, no. 4, July–August 2024, doi: 10.1109/TNSE.2024.3369045.
9. N.-P. Tran, **T. P. Truong**, Q. T. Do, N.-N. Dao, and S. Cho, “Adaptive Content Delivery and Bitrate Adaptation for QoE Improvement in IRS-Aided RSMA-Enabled IoMT Streaming Systems,” *Internet of Things*, vol. 25, no. 101145, April 2024.
10. **T. P. Truong**, V. D. Tuong, N.-N. Dao, and S. Cho, “FlyReflect: Joint Flying IRS Trajectory and Phase Shift Design using Deep Reinforcement Learning,” *IEEE Internet of Things Journal*, vol. 10, no. 5, pp. 4605–4620, March 2023, doi: 10.1109/JIOT.2022.3220581.
11. T.-V. Nguyen, **T. P. Truong**, T. M. T. Nguyen, W. Noh, and S. Cho, “Achievable Rate Analysis of Two-Hop Interference Channel with Coordinated IRS Relay,” *IEEE Transactions on Wireless Communications*, vol. 21, no. 9, pp. 7055–7071, September 2022, doi: 10.1109/TWC.2022.3155746.
12. **T. P. Truong**, N.-N. Dao, and S. Cho, “HAMEC-RSMA: Enhanced Aerial Computing Systems with Rate Splitting Multiple Access,” *IEEE Access*, vol. 10, pp. 52398–52409, May 2022, doi: 10.1109/ACCESS.2022.3174621.
13. V. D. Tuong, **T. P. Truong**, T.-V. Nguyen, W. Noh, and S. Cho, “Partial Computation Offloading in NOMA-Assisted Mobile Edge Computing Systems Using Deep Reinforcement Learning,” *IEEE Internet of Things Journal*, vol. 8, no. 17, pp. 13196–13208, September 2021, doi: 10.1109/JIOT.2021.3062481.

Selected Conference Papers

1. **T. P. Truong**, A.-T. Tran, V. D. Tuong, N.-N. Dao, and S. Cho, “NOMA-Enhanced Quantized Uplink Multi-user MIMO Communications,” in *Proc. of IEEE INFOCOM*, 2024. Acceptance rate: 19.6%.
2. J. Oh, D. Won, **T. P. Truong**, and S. Cho, “Communication-Efficient Federated Learning with Local-Reconstruction Error-Feedback-Based Rescaled 1-Bit Compressive Sensing,” in *Proc. of IEEE ICC*, Glasgow, Scotland, UK, May 2026.
3. **T. P. Truong**, T. M. T. Nguyen, T. V. Nguyen, N.-N. Dao, and S. Cho, “RSMA for Uplink MIMO Systems: DRL-based Achievable System Sum Rate Maximization,” in *Proc. of IEEE GLOBECOM Workshops*, 2023.
4. T. S. Do, **T. P. Truong**, H. H. Tran, K. N. T. Tran, M. C. Ho, A. T. Tran, V. N. Vu, and S. Cho, “Multi-Task Semantic Communication System for DNA Transmission,” in *Proc. of ICOIN*, Hanoi, Vietnam, January 2026. Best Paper Award.
5. **T. P. Truong**, T. S. Do, J. Kim, S. Choi, J. Oh, and S. Cho, “Efficient Offloading for Edge Computing in UAV-Assisted Maritime Networks,” in *Proc. of ICMIC*, Cebu, the Philippines, August 2025. Excellent Paper Award.
6. T. S. Do, **T. P. Truong**, Q. T. Do, D. Won, A. B. Wondmagegn, and S. Cho, “Super-Resolution Semantic Communication System for Satellite Image,” in *Proc. of ICAIIC*, Fukuoka, Japan, February 2025. Excellent Paper Award.
7. **T. P. Truong**, T.-H. Nguyen, A.-T. Tran, S. V.-T. Tran, V. D. Tuong, L. V. Nguyen, W. Na, L. Park, and S. Cho, “Deep Reinforcement Learning-Based Sum-Rate Maximization for Uplink Multi-user SIMO-RSMA Systems,” in *Proc. of ICIT*, Ho Chi Minh City, Vietnam, October 2023. Best Paper Award.
8. **T. P. Truong**, N.-P. Tran, C. M. Ho, T. T. H. Pham, A.-T. Tran, and S. Cho, “Multiple Access in HAP-Enabled Maritime IoT Networks: A Deep Reinforcement Learning Approach,” in *Proc. of ICMIC*, Jeju, South Korea, August 2023. Outstanding Paper Award.
9. T.-H. Nguyen, **T. P. Truong**, N.-N. Dao, W. Na, H. Park, and L. Park, “Deep Reinforcement Learning-based Partial Task Offloading in High Altitude Platform-aided Vehicular Networks,” in *Proc. of ICTC*, Jeju Island, South Korea, 2022. Best Paper Award.

References

Available upon request.